

Achille Thin

 Paris, France  thin.achille@gmail.com  +33 6 50 38 95 65  LinkedIn: achille-thin  Google Scholar

Research Scientist and Head of Data Science

Probabilistic machine learning, generative modelling, representation learning, and self-supervised learning

Research & Publications

Research profile and selected publications

Google Scholar

- Research at the intersection of variational inference, Monte Carlo methods, deep latent variable models, and representation learning, with a natural bridge toward embedding-based and self-supervised learning.
- *Monte Carlo Variational Auto-Encoders (ICML 2021)*: introduced a variational auto-encoding framework combining Monte Carlo objectives with deep generative modelling.
- *NEO: Non Equilibrium Sampling on the Orbit of a Deterministic Transform (NeurIPS 2021)*: developed efficient sampling procedures for high-dimensional probabilistic models and latent-variable learning.
- *BR-SNIS: Bias Reduced Self-Normalized Importance Sampling (NeurIPS 2022)* and *Differentiable Samplers for Deep Latent Variable Models (Philosophical Transactions of the Royal Society A, 2023)*: advanced reliable inference and differentiable sampling for deep models.
- Other scientific papers on applied systems: *A Validated Correction Method to Quantify Organic and Inorganic Carbon in Soils Using Rock-Eval Thermal Analysis (Biogeosciences, 2024)* and *First Modeling of Strongly Radiating WEST Plasmas with SOLEDGE-EIRENE (Nuclear Materials and Energy, 2019)*.

Education

PhD, Ecole polytechnique

Palaiseau, France

Statistics, Machine Learning, and Mathematical Modelling

September 2019–June 2022

- Thesis: *Novel Variational Approaches to Inference and Learning*.
- Research in variational inference, Monte Carlo methods, deep latent variable models, generative and representation learning.
- Publications at leading venues including ICML 2021 and NeurIPS 2021, with direct continuity toward modern embedding and self-supervised representation learning.

MSc in Statistical Science, University of Oxford

Oxford, United Kingdom

Statistics, Machine Learning, and Mathematical Modelling

September 2018–August 2019

- Thesis: Modelling and assessing the complexity of Campbell's monkey grammar with Bayesian nonparametric models.
- Advanced training in statistical inference, Bayesian modelling, machine learning, and stochastic processes.

Ecole polytechnique

Palaiseau, France

Engineering Degree (X2015)

September 2015–August 2018

- Rigorous training in mathematics, probability, statistics, computer science, and physics; ranked 23rd overall.

Experience

Head of Data Science, Nestlé Zone Europe

Paris, France

Consumer Modelling and Applied AI

March 2025–Present

- Lead a data science team working closely with stakeholders to shape and deliver consumer modelling initiatives.
- Design and prototype new machine learning models for stakeholder experimentation, including consumer embedding learning and self-supervised representation learning for richer behavioural modelling.
- Develop agentic AI frameworks and reusable workflows to accelerate experimentation, insight generation, and decision support.

Lead Data Scientist, Genesis

Paris, France

Soil Health and Environmental Modelling

2023–February 2025

- Defined the R&D strategy and led the development of soil health models based on physicochemical and biological indicators.
- Coordinated multidisciplinary projects to quantify the environmental impact of agroecology and conservation agriculture.
- Built modelling pipelines from multimodal scientific measurements, linking domain data to structured representations for downstream decision support.

Senior Lecturer and Researcher (Maître de Conférence), AgroParisTech

Saclay, France

Applied Mathematics and Computer Science Department

2022–2023

- Designed and taught MSc-level courses in machine learning, statistics, Bayesian methods, and applied data science.
- Conducted research and contributed to collaborative projects in applied mathematics, probabilistic modelling, and AI.

Research Intern, CEA

Cadarache, France

Fusion plasma modelling

2018

- Modelled plasma behaviour in fusion reactors to improve understanding of wall-plasma interactions.
- Developed a synthetic comparison method between simulations and experiments, contributing to a publication in *Nuclear Materials and Energy*.

Skills

 Python (PyTorch, Jax, scikit-learn), R, SQL, LaTeX; probabilistic machine learning, deep learning, generative modelling, representation learning, experimentation.

 Technical leadership, stakeholder collaboration, R&D strategy, scientific communication, teaching.

 French (native), English (fluent), Spanish (professional), Russian (basic).